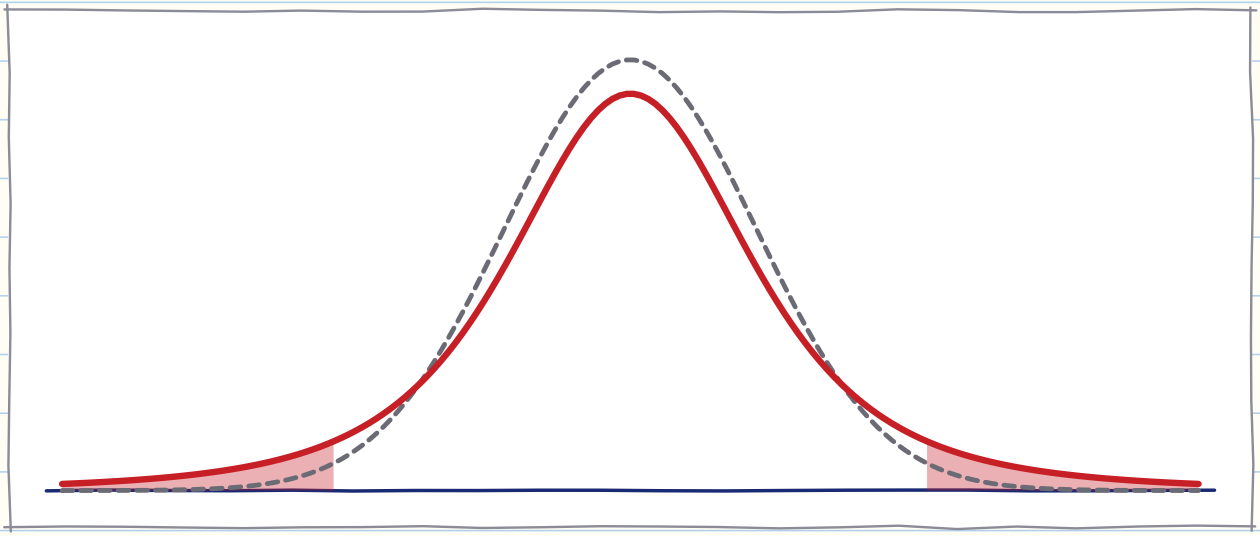
T-TEST

Handwritten-Style Study Notes

one-sample · pooled · Welch · paired

math intuition · logic · Excel · Python



normal (dashed) vs t (red): same bell, fatter tails

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statistics notebook

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How to read these notes

- **Blue ink** = main notes. **Red** = key terms and formulas.
- Green = tips. Purple boxes = intuition (the WHY). Orange = rules of thumb.
- Every test has: definition, formula, intuition, steps, worked numbers, Excel, Python.
- Section 19 = 10 practice problems. Section 20 = answer key (test, t, decision).

MASTER IDEA

Master skeleton, same for ALL four tests:

t = (what you observed − what H_0 says) / standard error

1. Why the t-Test Exists

the story behind Student's t

The problem we are solving

- You hold a **sample**. You want to say something about a **population mean**.
- Is the true mean fill 500 g? Does method A beat B? Did training help?
- If the population SD σ were KNOWN, we would use the z-test:

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

needs known σ

- Reality: σ is almost never known. All you have is the sample SD s.
- Swap σ → s and you add extra noise. s itself wobbles from sample to sample.
- So $(\bar{x} - \mu)/(s/\sqrt{n})$ is NOT normal. It follows the **t-distribution**.

HISTORY

Gosset story. William Sealy Gosset worked at the Guinness brewery in Dublin. He needed to judge small batches of barley and hops (n of 4, 5, 10). Normal tables lied for such tiny n. In 1908 he published the fix under the pen name 'Student'. That is why we say Student's t-test.

Intuition: z versus t

- t is a z-score with a **fuzzy denominator**. The ruler (s) is itself uncertain.
- Small n → s is unreliable → by luck s can be too small → t becomes huge.
- So extreme t values happen more often than extreme z values. Tails must be FATTER.
- Fatter tails → bigger critical value → harder to reject. Honest about doubt.
- More data → s → σ → the t curve slims down and becomes the normal curve.

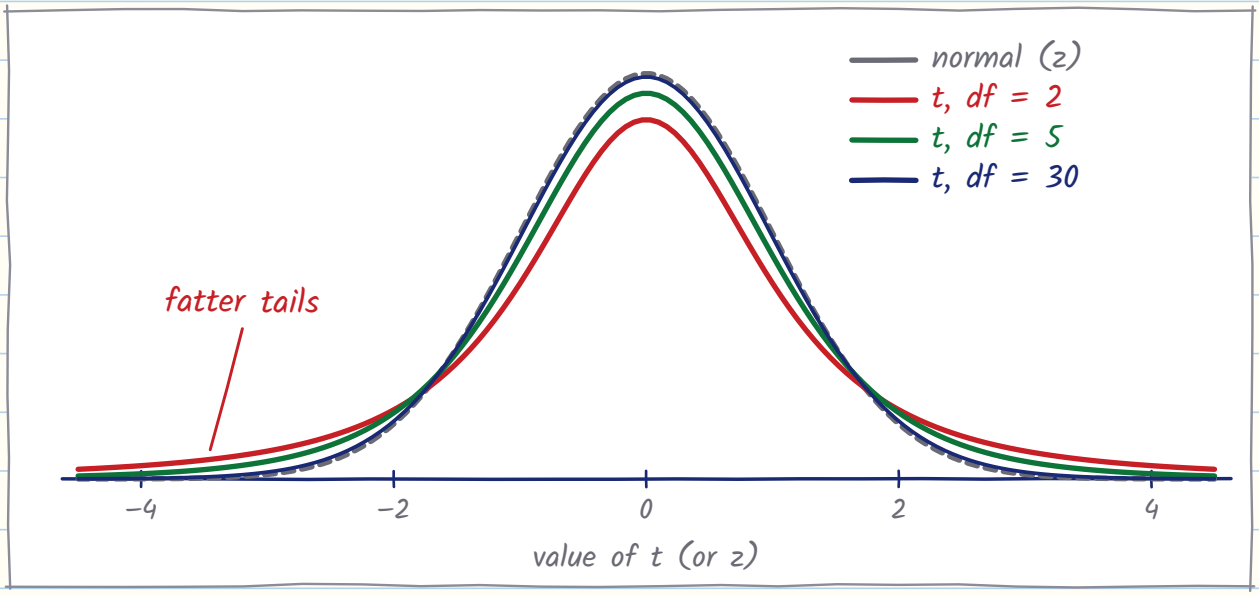


Fig 1: t curves vs the normal curve. Lower df = fatter tails.

Two-tailed 5% critical values

df	1	2	5	10	30	120	∞ (z)
t crit	12.706	4.303	2.571	2.228	2.042	1.980	1.960

- df = 1: you need |t| > 12.7 to be convinced. df = 30: |t| > 2.04 is enough.
- Big df and the table collapses onto the z value 1.96.

Where t-tests live: the family tree

- **One-sample**: one group vs a fixed number.
- **Independent two-sample**: two unrelated groups (pooled or Welch).
- **Paired**: the same subjects measured twice.

> **INTUITION**
>
> All four are one idea in four costumes: t = (estimate − null value) / SE.
>
> Only the estimate, the SE and the df change. Learn the skeleton once.

2. Building Blocks

mean, variance, SD, n−1, standard error, CLT

Sample mean = the balance point

$$\bar{x} = \frac{\Sigma\, x_i}{n}$$

- Add all values, divide by how many. It is where the data would balance on a seesaw.
- Property: deviations from the mean always sum to zero. $\Sigma(x - \bar{x}) = 0$.

Variance and SD = typical spread

$$s^2 = \frac{\Sigma\,(x_i - \bar{x})^2}{n - 1}\,,\quad s = \sqrt{(s^2)}$$

- Step 1: deviation = $x - \bar{x}$. Step 2: square it (kills the minus signs).
- Step 3: add the squares (this is SS, the sum of squares). Step 4: divide by $n - 1$.
- Step 5: square root to return to the original units → SD.

Mini example: 12, 15, 11, 14, 13

x	$x - \bar{x}$	$(x - \bar{x})^2$
12	−1	1
15	2	4
11	−2	4
14	1	1
13	0	0
sum = 65	0	SS = 10

$\bar{x} = 13$, $s^2 = 10/4 = 2.50$, $s = 1.581$

Why divide by n − 1? (Bessel's correction)

- We measure spread around $\bar{x}$, but $\bar{x}$ was fitted to THIS sample.
- $\bar{x}$ sits closer to the data than the unknown true μ does. So SS comes out too small.
- Dividing by n would underestimate σ^2 on average. Dividing by n − 1 fixes the bias.

INTUITION

Free-choice view. Deviations must add to 0. Pick n − 1 of them freely and the last one is FORCED. Only n − 1 pieces of independent information remain → degrees of freedom.

Standard error = spread of the MEAN

$$SE = s / \sqrt{n}$$

how far $\bar{x}$ typically lands from μ

- SD describes how spread out the DATA are. SE describes how spread out the MEAN is.
- Averaging cancels noise. Quadruple n → halve SE. (Square-root law.)

n	4	16	64	100	400
SE (if s = 10)	5.00	2.50	1.25	1.00	0.50

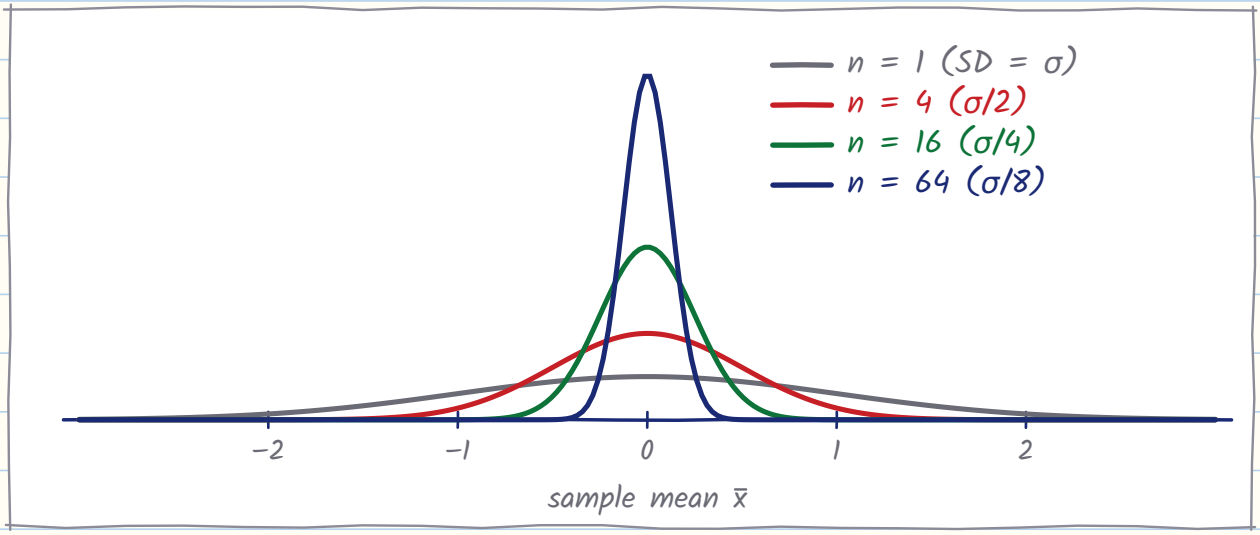


Fig 2: distribution of the sample mean tightens as n grows.

Central Limit Theorem (CLT)

- Whatever the data shape, the sample MEAN tends toward a normal curve as n grows.
- Rule of thumb: $n \geq 30$ is usually enough, unless data are very skewed.
- The t-test needs the MEAN to be roughly normal, not necessarily every single value.

Standardising = counting SEs

- t = how many standard errors my result sits away from the claim.
- t = 0 means 'exactly as claimed'. |t| near 2 or more means 'suspicious'.

RULE

Sanity feel: $|t| < 1$ is noise. $|t|$ around 2 is borderline. $|t| > 3$ is rarely luck.

3. Hypothesis Testing Framework

H_0, H_1, α, p-value, errors, power

The six steps (memorise)

1. State H_0 (nothing is going on) and H_1 (something is).
2. Pick α (usually 0.05) and decide one-tailed or two-tailed BEFORE looking.
3. Choose the test with the diagnostic flow (Section II).
4. Compute t and df.
5. Compare |t| with the critical value, or p with α.
6. Write the conclusion in plain words about the real situation.

Hypotheses

- **H_0 (null)**: no difference / no effect / the claim is true. Uses '=' (or ≤, ≥).
- **H_1 (alternative)**: what you suspect. Uses ≠ (two-tailed), > or < (one-tailed).
- We never prove H_0. We only ask: is the data too strange to live with H_0?

Significance level α

- α = the false-alarm rate you tolerate. At 0.05: 1 time in 20 you reject a true H_0.
- Choose α first. Never tune it after seeing the result.

The p-value

> **DEFINITION**
> p = probability of getting a result at least this extreme, ASSUMING H_0 is true. Small p means 'this data would be surprising in a world where H_0 holds'.

- $p < \alpha \rightarrow$ Reject H_0. $p \geq \alpha \rightarrow$ Fail to reject H_0.
- p is NOT 'the probability H_0 is true'. It is NOT the size of the effect either.

Critical-value method (same decision, different road)

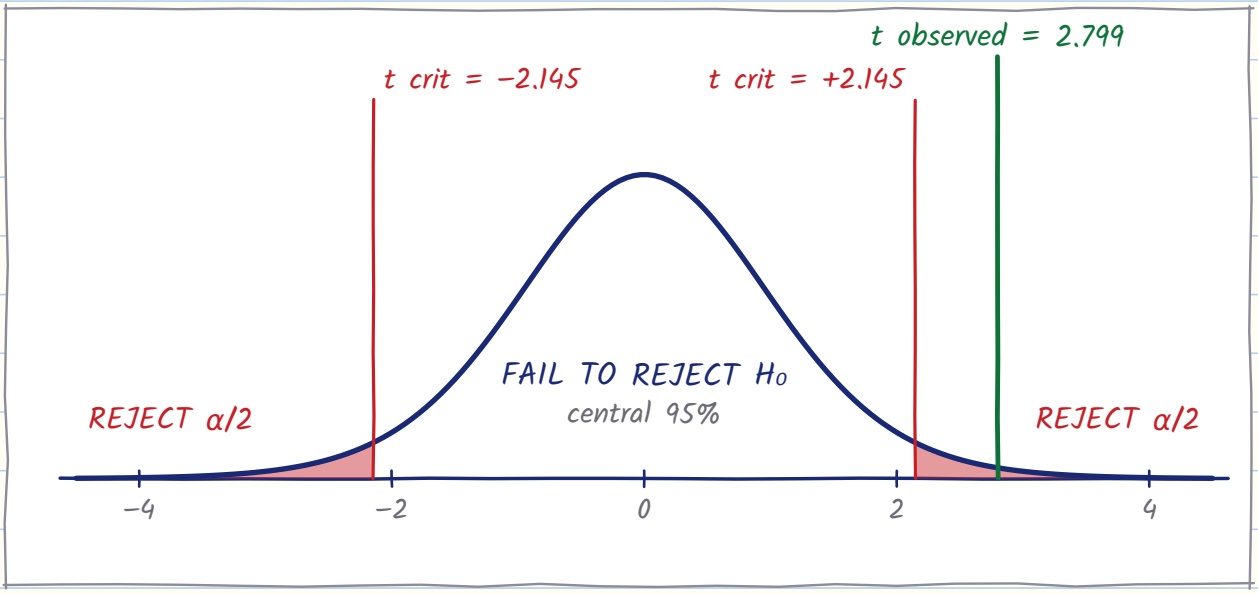


Fig 3: two-tailed test, df = 14, α = 0.05. t observed lands in the rejection zone.

- Find t-crit from the table for your df and α. If |t| > t-crit → reject H_0.
- Two-tailed: split α into two tails of α/2. One-tailed: all α in one tail.

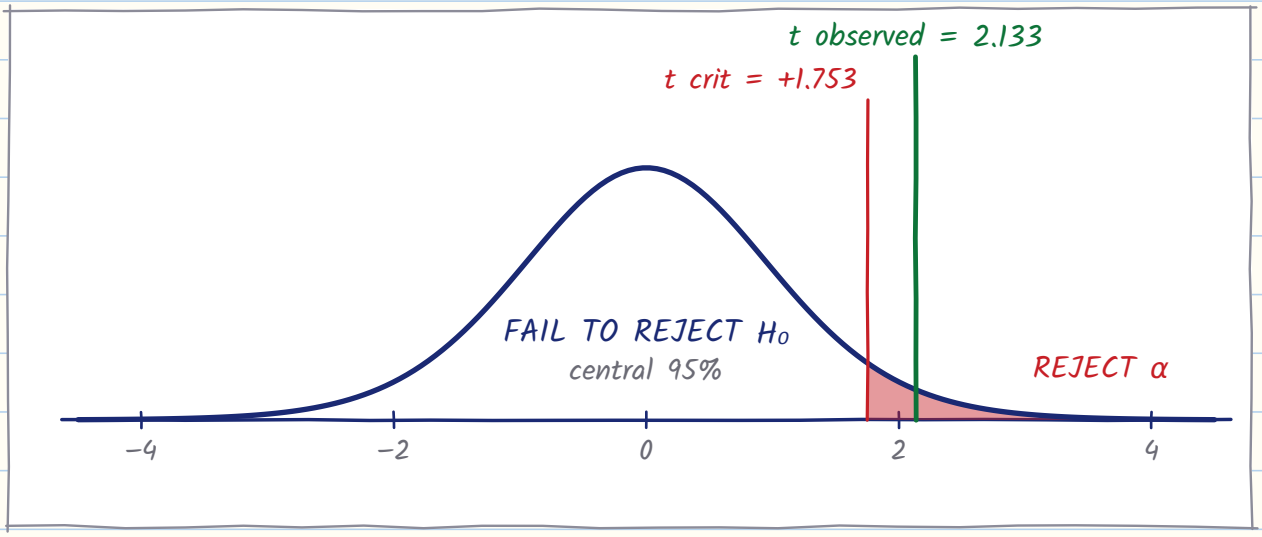


Fig 4: one-tailed (right) test, df = 15. All of α sits on one side.

Wording rules

- Say 'Reject H_0' or 'Fail to reject H_0'. Never say 'accept H_0'.
- Fail to reject = 'not enough evidence'. It is not 'proved equal'.

Type I and Type II errors

	H_0 really true	H_0 really false
Reject H_0	Type I error (α)	Correct: POWER (1−β)
Fail to reject	Correct (1−α)	Type II error (β)

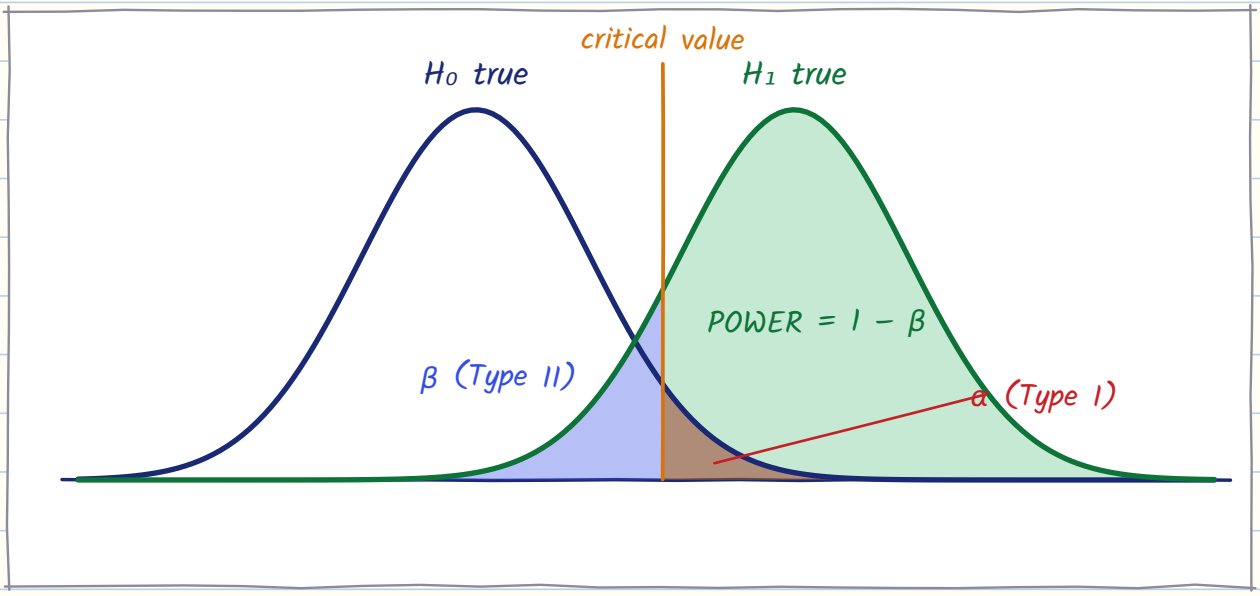


Fig 5: α, β and power. Moving the critical line trades one error for the other.

- Type I = false alarm (convict an innocent). Type II = miss (free a guilty).
- Power grows with: bigger true effect, bigger n, smaller noise (SD), larger α.

Confidence interval link

- 95% CI for the mean difference excludes 0 ⇔ two-tailed $p < 0.05$.

$$CI = \text{estimate} \pm t_{crit} \times SE$$

TIP

CI is better than p alone: it shows direction AND size AND uncertainty.

4. Anatomy of t and Degrees of Freedom

signal over noise, and what 'df' really means

t = signal / noise

$$t = \frac{\text{estimate} - \text{null value}}{\text{standard error}}$$

the universal template

- Numerator = **signal**: how far the data are from what H_0 claims.
- Denominator = **noise**: how much such a gap could appear by chance alone.
- Big $|t|$ = signal stands tall above noise. Small $|t|$ = could be noise.

Where the t-distribution comes from

$$t = \frac{Z}{\sqrt{(\chi^2_{df} / df)}}$$

- Top: Z is a standard normal. It is the clean $(\bar{x} - \mu)/(\sigma/\sqrt{n})$ part.
- Bottom: $\sqrt{(\text{chi-square}/df)}$ is the random wobble of the estimated SD, relative to σ.
- Wobble bigger when df is small. So the ratio is more extreme more often → fat tails.

INTUITION

When $df \to \infty$ the bottom $\to 1$ (estimated SD = true SD). Then $t = Z$. That is why the t table's last row equals the normal table.

Degrees of freedom (df) in plain words

- df = number of independent pieces of information left after estimating things.
- Example: three numbers must average 10. Choose two freely (say 8 and 14). The third is forced to 8. So $df = 3 - 1 = 2$.
- Every estimated mean 'uses up' one df.

Test	df	why
One-sample	$n - 1$	one mean estimated
Pooled	$n_1 + n_2 - 2$	two means estimated
Paired	$n - 1$	n differences, one mean
Welch	Satterthwaite (decimal)	estimated from s_1, s_2, n_1, n_2

- More df → thinner tails → smaller t-crit → easier to detect real effects.
- Paired: n counts PAIRS, not 2n observations.

RULE

Welch df always lies between $\min(n_1, n_2) - 1$ and $n_1 + n_2 - 2$. It equals the pooled df only when variances AND sample sizes match.

Reading a t-table

1. Pick the row = df.
2. Pick the column = α and tails (one-tailed 0.05 = two-tailed 0.10).
3. Compare |t observed| with that number.

- No row for your df (say 33)? Use the nearest SMALLER df. That is the safe side.

TIP

Excel/Python do this exactly: T.INV.2T(0.05, df) or stats.t.ppf(0.975, df).

5. Assumptions and How to Check Them

four assumptions, checks, and the escape routes

The four assumptions

1. **Independence**: observations do not influence each other (except inside a pair).
2. **Normality**: the data (or the differences) are roughly bell-shaped, OR n is large.
3. **No wild outliers**: one extreme point can drag the mean and inflate s.
4. **Equal variances**: ONLY for the pooled test. Welch drops this requirement.

How to check each one

Assumption	Quick check	Formal check
Independence	think about the design	none (design issue)
Normality	histogram / Q-Q plot	Shapiro-Wilk, $p > 0.05$ ok
Outliers	box plot, $\|z\| > 3$	Tukey fences 1.5 x IQR
Equal variance	SD ratio < 2	Levene, $p > 0.05$ ok

- Normality of what? One-sample: the data. Paired: the DIFFERENCES. Two-sample: each group.
- With $n \geq 30$ per group the CLT protects you from mild non-normality.
- Shapiro-Wilk with tiny n has little power; with huge n it flags trivial bumps. Use plots too.

What to do when an assumption fails

Problem	Escape route
Unequal variances	Welch t-test (safe default)
Skewed data, small n	log or sqrt transform, or rank test
One-sample / paired non-normal	Wilcoxon signed-rank test
Two groups non-normal	Mann-Whitney U test
Outlier from a typo	fix it; else report with and without

3 or more groups	ANOVA, NOT repeated t-tests

WATCH OUT

Repeating t-tests on many pairs inflates the false-alarm rate. Three groups = 3 pairs → about 14% chance of at least one false alarm at $\alpha = 0.05$. Use ANOVA.

assumption checks in Python

```
import numpy as np
from scipy import stats

x = np.array([78, 85, 82, 90, 74, 88, 81, 79])
y = np.array([72, 80, 75, 83, 70, 77, 74, 69])

# normality: p > 0.05 -> no evidence against normality
print('Shapiro x:', round(stats.shapiro(x).pvalue, 3))
print('Shapiro y:', round(stats.shapiro(y).pvalue, 3))

# equal variance: p > 0.05 -> pooled is acceptable
print('Levene  :', round(stats.levene(x, y).pvalue, 3))
print('SD ratio:', round(x.std(ddof=1) / y.std(ddof=1), 2))

# outliers: z-scores beyond +-3
z = (x - x.mean()) / x.std(ddof=1)
print('max |z| :', round(abs(z).max(), 2))

# Q-Q plot (needs matplotlib)
# stats.probplot(x, plot=plt); plt.show()
# output:
# Shapiro x: 0.956
# Shapiro y: 0.836
# Levene  : 0.801
# SD ratio: 1.1
# max |z| : 1.52
```

TIP

If Levene is borderline, do not agonise. Run Welch. It loses almost nothing when variances are equal and rescues you when they are not.

6. CORE DEFINITIONS: The Four t-Tests

the four tests in plain English, then in symbols

PLAIN ENGLISH

One-sample: is MY group different from a benchmark? Pooled: do two DIFFERENT groups differ (similar spread)? Welch: same, but spreads differ. Paired: did the SAME subjects change?

1) One-sample t-test

- **Definition**: tests whether one population mean equals a stated value μ_0.
- H_0: $\mu = \mu_0$. Example: 'cereal boxes average 500 g'.

$$t = \frac{\bar{x} - \mu_0}{s \,/\, \sqrt{n}}, \quad df = n - 1$$

2) Independent two-sample, equal variance (pooled)

- **Definition**: compares the means of two unrelated groups, assuming $\sigma_1^2 = \sigma_2^2$.
- H_0: $\mu_1 = \mu_2$. Blends both samples' variances into ONE estimate s_p^2.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{[\, s_p^2 \,(1/n_1 + 1/n_2)\,]}}, \quad df = n_1 + n_2 - 2$$

$$s_p^2 = [\,(n_1-1)s_1^2 + (n_2-1)s_2^2\,] \,/\, (n_1 + n_2 - 2)$$

3) Independent two-sample, unequal variance (Welch)

- **Definition**: same question as pooled, but each group keeps its OWN variance.
- H_0: $\mu_1 = \mu_2$. No equal-variance assumption. df is adjusted (Satterthwaite).

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{(s_1^2/n_1 + s_2^2/n_2)}}$$

4) Paired t-test

- **Definition**: tests whether the MEAN DIFFERENCE within matched pairs is zero.
- H_0: $\mu_d = 0$. Compute $d = x_1 - x_2$ per pair, then do a one-sample test on d.

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}, \quad df = n - 1 \ (n = \text{number of PAIRS})$$

Side-by-side summary

Test	Data layout	H_0	df
One-sample	1 column + μ_0	$\mu = \mu_0$	$n - 1$
Pooled	2 unrelated columns	$\mu_1 = \mu_2$	n_1+n_2-2
Welch	2 unrelated columns	$\mu_1 = \mu_2$	Satterthwaite
Paired	2 linked columns	$\mu_d = 0$	$n - 1$

TIP

Hidden connection: the paired t-test IS a one-sample t-test applied to the differences, with $\mu_0 = 0$.

7. One-Sample t-Test (Deep Dive)

one group against a benchmark

Idea and intuition

- Claim: the true mean is μ_0. Sample says $\bar{x}$. Is the gap just luck?
- Measure the gap in units of SE. That is t.

$$t = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}, \quad df = n - 1$$

INTUITION

Gap far bigger than the SE → hard to blame luck. Gap similar to the SE → luck is a perfectly good explanation.

Steps

1. H_0: $\mu = \mu_0$. H_1: $\mu \neq \mu_0$ (or $>$ / $<$).
2. Compute $\bar{x}$, s, $SE = s/\sqrt{n}$.
3. $t = (\bar{x} - \mu_0)/SE$, $df = n - 1$.
4. Compare with t-crit or find p. Decide. Write the sentence.

Worked example: tube-light lifetimes

Maker claims mean life = 1000 hours. Test 9 lights: 985, 1012, 968, 990, 1005, 975, 992, 980, 998.

Is the mean different from 1000 h? $\alpha = 0.05$, two-tailed.

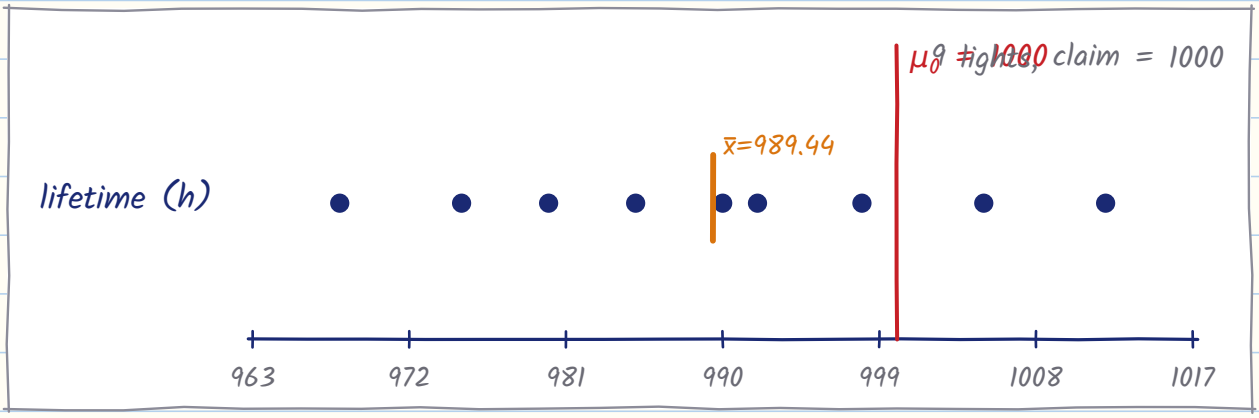


Fig 6: data cluster below the claim, but is it more than noise?

x	$x - \bar{x}$	$(x - \bar{x})^2$
985	−4.44	19.75
1012	22.56	508.75
968	−21.44	459.86
990	0.56	0.31
1005	15.56	241.98
975	−14.44	208.64
992	2.56	6.53
980	−9.44	89.20
998	8.56	73.20
$\Sigma x = 8905$	≈ 0	$SS = 1608.22$

1. H_0: $\mu = 1000$. H_1: $\mu \neq 1000$.
2. $\bar{x} = 8905/9 = 989.444$
3. $s^2 = SS/(n-1) = 1608.22/8 = 201.03$, $s = 14.178$
4. $SE = s/\sqrt{n} = 14.178/3 = 4.726$
5. $t = (989.44 - 1000)/4.726 = \mathbf{-2.233}$
6. $df = 8$. t-crit (two-tailed, 5%) = ±2.306. $p = \mathbf{0.0560}$
7. $|t| = 2.233 < 2.306$ (and $p > 0.05$) → **FAIL TO REJECT H_0.**
8. 95% CI for $\mu = 989.44 \pm 2.306 \times 4.726 = (978.5, 1000.3)$. It contains 1000 ✓.

SENTENCE

Conclusion: at 5% there is not enough evidence that mean life differs from 1000 h ($t(8) = -2.23$, $p = 0.056$). The result is borderline: it WOULD be significant at 10%. Never write 'proved the claim true'.

Excel (data in A2:A10)

```
B1  =AVERAGE(A2:A10)                    -> 989.444
B2  =STDEV.S(A2:A10)                    -> 14.178
B3  =COUNT(A2:A10)                      -> 9
B4  =B2/SQRT(B3)                        -> 4.726   (SE)
B5  =(B1-1000)/B4                       -> -2.233   (t)
B6  =T.INV.2T(0.05,B3-1)                -> 2.306    (t crit)
B7  =T.DIST.2T(ABS(B5),B3-1)            -> 0.0560   (p)
B8  =B1-B6*B4   and   =B1+B6*B4         -> CI ends
```

Python

```python
import numpy as np
from scipy import stats

life = [985, 1012, 968, 990, 1005, 975, 992, 980, 998]
res = stats.ttest_1samp(life, popmean=1000)
print('t  =', round(res.statistic, 3))
print('df =', res.df)
print('p  =', round(res.pvalue, 4))
ci = res.confidence_interval(0.95)
print('CI =', round(ci.low, 1), round(ci.high, 1))

# one-tailed version: is mean LESS than 1000?
r1 = stats.ttest_1samp(life, 1000, alternative='less')
print('one-tailed p =', round(r1.pvalue, 4))
# output:
# t  = -2.233
# df = 8
# p  = 0.056
# CI = 978.5 1000.3
# one-tailed p = 0.028
```

WATCH OUT

Notice: one-tailed p = half the two-tailed p when the sample is on the predicted side. Decide tails BEFORE seeing data, or you are cheating.

8. Independent t-Test, Equal Variance

two unrelated groups, assuming equal spread

Idea

- Two separate groups. Compare their means. Assume both come from populations with the SAME variance σ^2.
- If true, both samples are estimating the same σ^2. So combine them for a better estimate.

Step A: pooled variance = weighted average of s_1^2 and s_2^2

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1 + n_2 - 2}$$

- Weights are the df of each group. Bigger group → more say.
- s_p^2 always lies between s_1^2 and s_2^2.

TIP

Equal n? Then s_p^2 is just the plain average of the two variances.

Step B: where the standard error comes from

- Each sample mean has variance σ^2/n. The two groups are independent.
- Variances ADD, even when you subtract the means:

$$Var(\bar{x}_1 - \bar{x}_2) = \sigma^2/n_1 + \sigma^2/n_2 = \sigma^2(1/n_1 + 1/n_2)$$

- Replace σ^2 with s_p^2 and take the root:

$$SE = \sqrt{[\, s_p^2 \, (1/n_1 + 1/n_2) \,]}$$

INTUITION

Subtracting means does NOT subtract noise. Noise from both groups piles up. That is why SE has both $1/n_1$ and $1/n_2$ with a plus sign.

Step C: t and df

$$t = \frac{\bar{x}_1 - \bar{x}_2}{SE}, \quad df = n_1 + n_2 - 2$$

- Two means estimated → lose two df.

When to use pooled

- Design looks balanced and SD ratio < 2 (variance ratio < 4).
- Levene's test not significant, and you have a reason to expect equal spread.
- Benefit: more df, a hair more power. Risk: wrong if variances differ a lot (Section 9).

Mini example from summary statistics

Group 1: $n = 10$, mean 50, SD 5. Group 2: $n = 12$, mean 46, SD 6. $\alpha = 0.05$ two-tailed.

1. SD ratio = 6/5 = 1.2 < 2 → pooled is fine.
2. $s_p^2 = (9 \times 25 + 11 \times 36)/20 = (225 + 396)/20 = \mathbf{31.05}$
3. $SE = \sqrt{[31.05 \times (1/10 + 1/12)]} = \sqrt{[31.05 \times 0.18333]} = 2.386$
4. $t = (50 - 46)/2.386 = \mathbf{1.677}$, $df = 20$
5. t-crit = ±2.086, $p = 0.109$ → **FAIL TO REJECT H_0.**

summary-stat version

```
from scipy import stats
r = stats.ttest_ind_from_stats(mean1=50, std1=5, nobs1=10,
                               mean2=46, std2=6, nobs2=12,
                               equal_var=True)
print(round(r.statistic, 3), round(r.pvalue, 3))
# output:
# 1.677 0.109
```

NOTE

A 4-unit gap looks big, but SE is 2.39. The gap is only 1.68 SEs wide. Not enough.

9. Independent t-Test, Unequal Variance

Welch: two groups, each with its own spread

Idea

- Do NOT pretend the variances match. Let each group bring its own s^2.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\left(s_1^2/n_1 + s_2^2/n_2 \right)}}$$

- Same numerator as pooled. Only the standard error (and df) change.

Degrees of freedom (Welch-Satterthwaite)

$$df = \frac{\left(s_1^2/n_1 + s_2^2/n_2 \right)^2}{(s_1^2/n_1)^2/(n_1-1) + (s_2^2/n_2)^2/(n_2-1)}$$

- Ugly, but only a calculator button. Result is a decimal. Do NOT round it for software.

INTUITION

Intuition: the noisier, smaller group controls the answer. If one group is erratic and tiny, df collapses toward that group's $n - 1$, because that is where the uncertainty lives.

- Bounds: $\min(n_1, n_2) - 1 \leq df \leq n_1 + n_2 - 2$.

Why not always pool? A cautionary example

Server response time (ms). Server A: 10 stable requests. Server B: only 6 erratic requests.

A = 98, 104, 101, 95, 103, 99, 107, 100, 96, 102

B = 92, 140, 118, 125, 96, 131

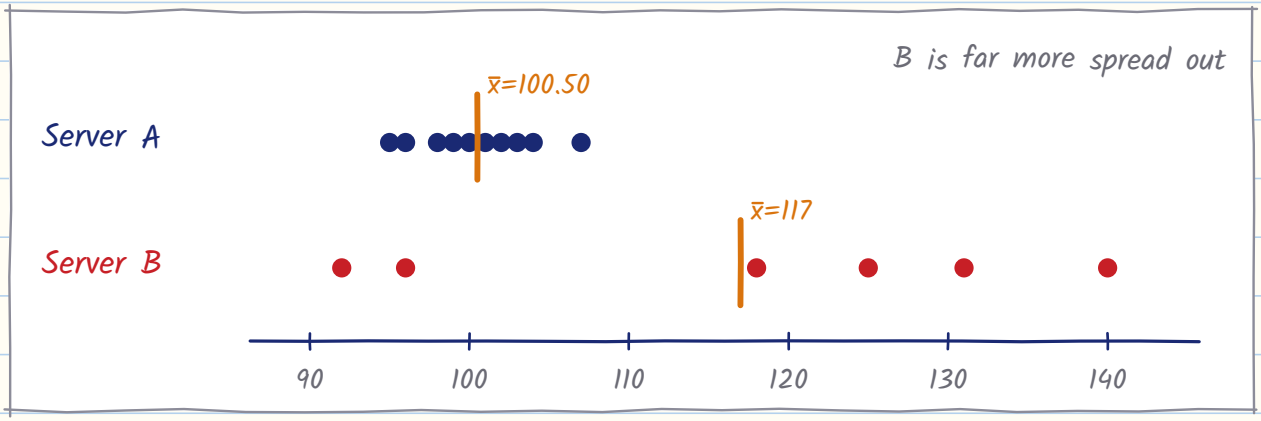


Fig 7: same-ish centre story, but B's spread dwarfs A's.

Welch by hand

1. $\bar{x}_A = 100.5$, $s_A^2 = 13.61$ (SD 3.69); $\bar{x}_B = 117.0$, $s_B^2 = 371.20$ (SD 19.27)
2. SD ratio = 19.27/3.69 = 5.22 (> 2 → use Welch)
3. $s_A^2/n_A = 13.61/10 = 1.361$; $s_B^2/n_B = 371.20/6 = 61.867$
4. SE = $\sqrt{(1.361 + 61.867)} = \sqrt{63.228} = 7.952$
5. $t = (100.5 - 117.0)/7.952 = \mathbf{-2.075}$
6. $df = (63.228)^2 / [\ 1.361^2/9 + 61.867^2/5\ ] = \mathbf{5.22}$
7. t-crit(df 5.22) = ±2.538; p = 0.090 → **FAIL TO REJECT**

Naive pooled on the SAME data

- $s_p^2 = 141.3$, SE = 6.14, t = −2.688, df = 14, p = **0.018** → would REJECT.

WATCH OUT

Pooled is fooled. Its variance estimate is dominated by the big, calm group A, so SE is too small and t too big. When the SMALLER group has the LARGER variance, pooled gives too many false alarms. Welch correctly says 'not enough evidence'.

Excel

```
=T.TEST(A2:A11,B2:B7,2,3)    <- type 3 = Welch, 2-tailed -> 0.0903
=T.TEST(A2:A11,B2:B7,2,2)    <- type 2 = pooled, 2-tailed -> 0.0177
ToolPak: 't-Test: Two-Sample Assuming UNEQUAL Variances'  (shows df)
```

Python

```
from scipy import stats
A = [98, 104, 101, 95, 103, 99, 107, 100, 96, 102]
B = [92, 140, 118, 125, 96, 131]
w = stats.ttest_ind(A, B, equal_var=False)   # Welch
```

```
p = stats.ttest_ind(A, B, equal_var=True)    # pooled
print('Welch :', round(w.statistic, 3), round(w.df, 2), round(w.pvalue, 3))
print('Pooled:', round(p.statistic, 3), p.df, round(p.pvalue, 3))
# output:
# Welch : -2.075 5.22 0.09
# Pooled: -2.688 14.0 0.018
```

TIP

Default habit: use Welch unless you have a strong reason not to. R's t.test() does this by default. When variances ARE equal, Welch ≈ pooled, so you lose almost nothing.

10. Paired t-Test (Deep Dive)

same subjects, two readings: test the differences

Idea

- Before/after on the SAME people, twins, left vs right hand, matched pairs.
- Step 1: for each pair compute d = after − before (keep the sign!).
- Step 2: run a one-sample t-test on the d values against 0.

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}, \quad df = n - 1$$

Why pairing is powerful (the math)

$$Var(x_1 - x_2) = \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$$

- ρ = correlation between the two readings of the same subject.
- People who score high before also score high after, so $\rho > 0$.
- Positive ρ SUBTRACTS noise. Differences are far steadier than raw scores.

> **INTUITION**
> Each person is their own control. Person-to-person differences (height, skill, baseline) cancel out. You only see the change.

- If $\rho = 0$, pairing buys nothing and costs you df. If $\rho < 0$, pairing hurts. Rare.

Worked example: typing speed before/after training

7 employees, words per minute. Did training change speed? $\alpha = 0.05$, two-tailed.

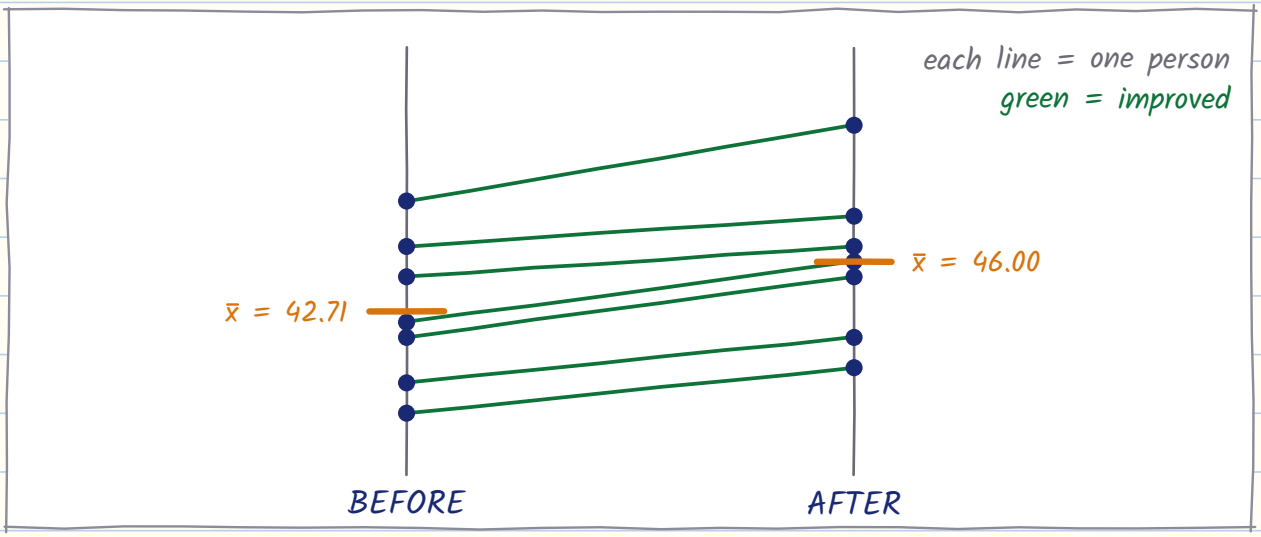


Fig 8: nearly every line climbs; pairing shows it clearly.

person	before	after	d = after−before	d−d̄	(d−d̄)²
1	42	46	4	0.714	0.510
2	38	41	3	−0.286	0.082
3	45	47	2	−1.286	1.653
4	50	55	5	1.714	2.939
5	36	39	3	−0.286	0.082
6	41	45	4	0.714	0.510
7	47	49	2	−1.286	1.653
			Σd = 23		SS = 7.429

1. H_0: $\mu_d = 0$. H_1: $\mu_d \neq 0$.
2. $\bar{d} = 23/7 = 3.2857$
3. $s_d^2 = 7.429/6 = 1.2381$, $s_d = 1.1127$
4. $SE = 1.1127/\sqrt{7} = 0.4206$
5. $t = 3.2857/0.4206 = \mathbf{7.813}$, $df = 6$
6. $t\text{-crit} = \pm 2.447$, $p = 0.00023 \rightarrow$ **REJECT H_0**. Training raised speed.
7. 95% CI for $\mu_d = 3.29 \pm 2.447 \times 0.421 = (2.26, 4.31)$ words/min.

What if we wrongly ran an independent test?

- Independent t on the same numbers: $t = 1.203$, $p = 0.252 \rightarrow$ **not significant**.
- Same data, opposite decision. The big person-to-person spread (36 to 55 wpm) drowned the steady +3.3 gain.

WATCH OUT

Never ignore pairing. A paired design analysed as independent wastes power. An independent design analysed as paired is plain wrong.

Excel (before in A2:A8, after in B2:B8)

```
C2  =B2-A2    (fill down to C8)            <- differences d
E1  =AVERAGE(C2:C8)                        -> 3.2857
E2  =STDEV.S(C2:C8)                        -> 1.1127
E3  =E1/(E2/SQRT(COUNT(C2:C8)))            -> 7.813   (t)
E4  =T.DIST.2T(ABS(E3),COUNT(C2:C8)-1)     -> 0.00023
E5  =T.TEST(A2:A8,B2:B8,2,1)               -> same p (type 1 = paired)
```

Python

```python
from scipy import stats
before = [42, 38, 45, 50, 36, 41, 47]
after  = [46, 41, 47, 55, 39, 45, 49]
r = stats.ttest_rel(after, before)
print(round(r.statistic, 3), r.df, round(r.pvalue, 5))

# identical: one-sample test on the differences
d = [a - b for a, b in zip(after, before)]
print(round(stats.ttest_1samp(d, 0).statistic, 3))

# the WRONG test (independent) for comparison
w = stats.ttest_ind(after, before)
print(round(w.statistic, 3), round(w.pvalue, 3))
# output:
# 7.813 6 0.00023
# 7.813
# 1.203 0.252
```

II. DIAGNOSTIC FLOW: Which Test?

quick, punchy rules: pick the right test in 10 seconds

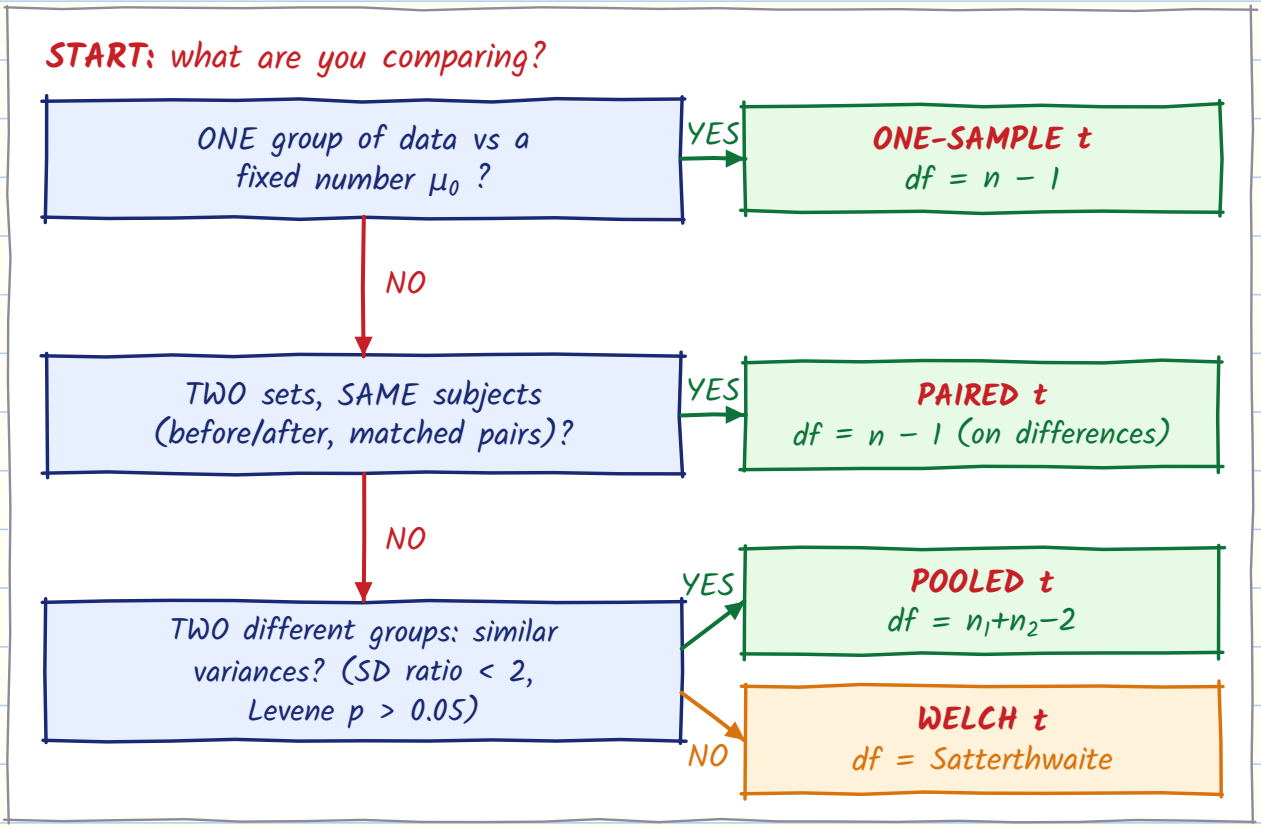


Fig 9: the decision tree. Answer top to bottom.

Punchy rules

- Compare to a fixed number? **One-sample**.
- Same subjects measured twice (or matched)? **Paired**.
- Different subjects in each group? **Independent**. Then check variances.
- SD ratio < 2 and balanced design? **Pooled**. Otherwise or unsure? **Welch**.
- 'Greater / less / improved / reduced' in the claim? **One-tailed**. 'Different / changed'? **Two-tailed**.
- Only summary stats (mean, SD, n)? Same formulas. Use the summary-stat versions.

RULE

When two tests seem possible, ask: 'Can I pair row 1 of group A with row 1 of group B for a real reason?' No real link = independent.

Scenario drill

Scenario	Test
Mean of 25 packets vs label weight	One-sample
Same 12 patients, BP before and after drug	Paired
Identical twins, one gets tutoring	Paired
Control vs treatment, random assignment	Independent
Two machines, SD 2 g vs SD 9 g	Welch
Left-hand vs right-hand speed, same people	Paired
Men vs women salaries, similar spread	Pooled (or Welch)
Before/after, but DIFFERENT people each time	Independent

Not a t-test situation

Situation	Use instead
3+ group means	ANOVA
Proportions / percentages	z-test for proportions, chi-square
Small n, heavy skew, outliers	Wilcoxon / Mann-Whitney
Known population σ, any n	z-test
Compare variances	F-test, Levene

12. COMPREHENSIVE NUMERICAL WORKING

one full scenario: every step, Excel, Python

Scenario

A coaching institute tests two teaching methods on 8 different students each. Method A is interactive, Method B is lecture. Marks out of 100.

Question: is there a significant difference in mean marks? $\alpha = 0.05$, two-tailed.

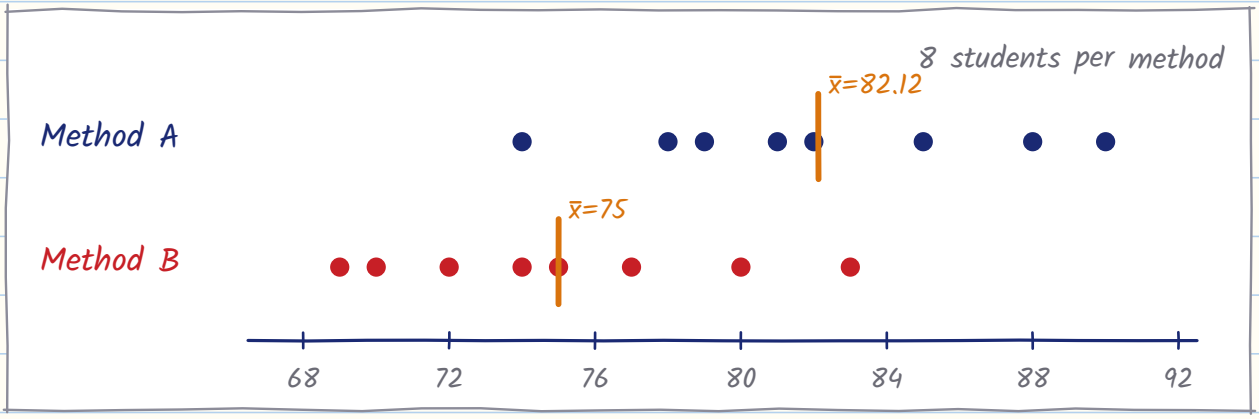


Fig 10: the A cloud sits to the right of B. Significant or luck?

Student	1	2	3	4	5	6	7	8
Method A	78	85	82	90	74	88	81	79
Method B	72	80	75	83	70	77	74	69

Test choice

- Different students in each group → **independent** (not paired).
- $SD_A = 5.33$, $SD_B = 4.84$ → ratio 1.10 < 2 → **pooled / equal variance**.

> **NOTE**
> Since $n_1 = n_2$ and SDs match, Welch would give the identical t and nearly the same p. Pooled is the clean textbook route.

Step 1: hypotheses

- H_0: $\mu_A = \mu_B$ H_1: $\mu_A \neq \mu_B$ (two-tailed, $\alpha = 0.05$)

Step 2: means

- $\bar{x}_A = 657/8 = \mathbf{82.125}$ $\bar{x}_B = 600/8 = \mathbf{75.000}$

Step 3: squared deviations

A	A−x̄	(A−x̄)²	B	B−x̄	(B−x̄)²
78	−4.125	17.016	72	−3	9
85	2.875	8.266	80	5	25
82	−0.125	0.016	75	0	0
90	7.875	62.016	83	8	64
74	−8.125	66.016	70	−5	25
88	5.875	34.516	77	2	4
81	−1.125	1.266	74	−1	1
79	−3.125	9.766	69	−6	36
		$SS_A = 198.875$			$SS_B = 164$

Step 4: variances

- $s_A^2 = 198.875/7 = \mathbf{28.411}$ $s_B^2 = 164/7 = \mathbf{23.429}$

Step 5: pooled variance

- $s_p^2 = (198.875 + 164)/14 = 362.875/14 = \mathbf{25.9196}$ ($s_p = 5.091$)

Step 6: standard error

- $SE = \sqrt{[\ 25.920 \times (1/8 + 1/8)\ ]} = \sqrt{6.4799} = \mathbf{2.5456}$

Step 7: t-statistic

$$t = \frac{82.125 - 75.000}{2.5456} = \mathbf{2.799}$$

Step 8: decision

- df = 8 + 8 − 2 = **14**. t-crit (two-tailed, 5%) = **±2.145**.
- |2.799| > 2.145 → **REJECT H_0**. p = **0.0142** < 0.05 ✓

Step 9: confidence interval and effect size

- 95% CI for $\mu_A - \mu_B$ = 7.125 ± 2.145 × 2.5456 = **(1.67, 12.58)** → excludes 0 ✓
- Cohen's d = 7.125 / 5.091 = **1.40** (large effect, > 0.8)

CONCLUSION SENTENCE

Method A gives significantly higher marks than Method B: t(14) = 2.80, p = 0.014. The mean advantage is 7.1 marks (95% CI 1.7 to 12.6).

Excel: exact formulas

Put Method A in A2:A9 and Method B in B2:B9.

Excel cells

```
E2  =AVERAGE(A2:A9)                                -> 82.125
E3  =AVERAGE(B2:B9)                                -> 75.000
E4  =VAR.S(A2:A9)                                  -> 28.4107
E5  =VAR.S(B2:B9)                                  -> 23.4286
E6  =(7*E4+7*E5)/14                                -> 25.9196  (pooled var)
E7  =SQRT(E6*(1/8+1/8))                            -> 2.5456  (SE)
E8  =(E2-E3)/E7                                    -> 2.7990  (t)
E9  =8+8-2                                         -> 14   (df)
E10 =T.INV.2T(0.05,E9)                             -> 2.1448  (t crit)
E11 =T.DIST.2T(ABS(E8),E9)                         -> 0.0142  (p)
E12 =T.TEST(A2:A9,B2:B9,2,2)                       -> 0.0142  (direct p)
E13 =IF(E11<0.05,"Reject H0","Fail to reject")  -> Reject H0
E14 =(E2-E3)-E10*E7   |   E15 =(E2-E3)+E10*E7 -> CI 1.665, 12.585
E16 =(E2-E3)/SQRT(E6)                              -> 1.399  (Cohen d)
```

WATCH OUT

T.TEST returns ONLY the p-value. It never shows t. To get t you must build it as in E8, or use the ToolPak.

Excel ToolPak path (click by click)

1. File → Options → Add-ins → Manage: Excel Add-ins → Go → tick **Analysis ToolPak** → OK.
2. Data tab → **Data Analysis** (far right).
3. Select **t-Test: Two-Sample Assuming Equal Variances** → OK.
4. Variable 1 Range = A1:A9. Variable 2 Range = B1:B9.
5. Hypothesized Mean Difference = 0. Tick Labels. Alpha = 0.05.
6. Pick an Output Range cell (say G1) → OK.
7. Read the table: t Stat, P(T<=t) two-tail, t Critical two-tail, df.

ToolPak output	Value
Pooled Variance	25.9196
df	14
t Stat	2.7990
P(T<=t) two-tail	0.0142
t Critical two-tail	2.1448

> **TIP**
> The ToolPak also offers 'Unequal Variances' and 'Paired Two Sample for Means'. Same dialog, different test.

Python (SciPy)

```python
import numpy as np
from scipy import stats

A = [78, 85, 82, 90, 74, 88, 81, 79]
B = [72, 80, 75, 83, 70, 77, 74, 69]
alpha = 0.05

res = stats.ttest_ind(A, B, equal_var=True)      # pooled
print('t  =', round(res.statistic, 3))
print('df =', res.df)
print('p  =', round(res.pvalue, 4))
print('Reject H0' if res.pvalue < alpha else 'Fail to reject H0')

ci = res.confidence_interval(0.95)
print('CI =', round(ci.low, 2), round(ci.high, 2))
```

```
sp = np.sqrt((np.var(A, ddof=1) + np.var(B, ddof=1)) / 2)
print('Cohen d =', round((np.mean(A) - np.mean(B)) / sp, 2))

w = stats.ttest_ind(A, B, equal_var=False)       # Welch check
print('Welch df =', round(w.df, 2), ' p =', round(w.pvalue, 4))
# output:
# t  = 2.799
# df = 14.0
# p  = 0.0142
# Reject H0
# CI = 1.67 12.58
# Cohen d = 1.4
# Welch df = 13.87  p = 0.0143
```

NOTE

Welch gives the same t but df = 13.87 and p = 0.0143. Same decision.

That is what you expect when variances and n are balanced.

13. Tails, Effect Size and Power

one vs two tails, how BIG is the effect, and how much data you need

One-tailed vs two-tailed

	Two-tailed	One-tailed
H_1	$\mu \neq \mu_0$ (any change)	$\mu > \mu_0$ or $\mu < \mu_0$
Keywords	different, changed, differs	greater, less, improved, reduced
α placement	$\alpha/2$ in each tail	all α in one tail
t-crit (df 14, 5%)	±2.145	+1.761 (or −)
p-value	p	$p/2$ (if sign matches H_1)

- Decide the direction BEFORE collecting data, from theory, not from the sample.
- One-tailed has more power in its direction and ZERO power in the other.
- If the sample goes the 'wrong' way, you cannot reject, however large the gap.

> **RULE**
> Safe default: two-tailed. Use one-tailed only when a result in the other direction would be treated exactly like 'no effect'.

Excel one-tailed p

```
Right tail (H1: mean >):   =T.DIST.RT(t, df)
Left tail  (H1: mean <):   =T.DIST(t, df, TRUE)
T.TEST(a1, a2, 1, type) -> one-tailed p (tails argument = 1)
```

Effect size: how BIG, not just 'is it real'

- p answers 'is there an effect?'. Effect size answers 'how large is it?'.

Test	Cohen's d
One-sample	$(\bar{x} - \mu_0)/s$
Independent	$(\bar{x}_1 - \bar{x}_2)/s_p$ (Welch: use average SD)
Paired	$\bar{d}/s_d$ (called d_z)

|d|	0.2	0.5	0.8
label	small	medium	large

- Small samples overstate d. Hedges' $g = d \times [1 - 3/(4 \cdot df - 1)]$. Scenario of Section 12: $g = 1.40 \times (1 - 3/55) = 1.32$.

INTUITION

Link between t and d (equal n, two groups): $t = d \times \sqrt{(n/2)}$. Check: $1.40 \times \sqrt{4} = 2.80$ ✓. One-sample/paired: $t = d \times \sqrt{n}$.

- Huge n makes trivial effects 'significant': $d = 0.05$ with $n = 10{,}000$ per group gives $t \approx 3.54$, yet the effect is tiny. Statistical ≠ practical significance.

Power and sample size

- Power = $1 - \beta$ = chance of detecting a real effect. Aim for 0.80 or more.
- Raise power by: more n, bigger effect, lower noise, larger α, paired design, one tail.

$$n \text{ per group} \approx 2\,(z_{1-\alpha/2} + z_{1-\beta})^2 \,/\, d^2$$

two-sample, equal n

effect d	0.2 (small)	0.5 (medium)	0.8 (large)
n per group (normal approx)	393	63	25

($\alpha = 0.05$ two-tailed, power 0.80. Exact t-based answers run 1 to 2 higher.)

power / sample size in SciPy

```
import numpy as np
from scipy import stats

def power_ind(d, n, alpha=0.05):      # equal n, two-tailed
    df = 2 * n - 2
    nc = d * np.sqrt(n / 2)           # noncentrality
    tc = stats.t.ppf(1 - alpha / 2, df)
    return stats.nct.sf(tc, df, nc) + stats.nct.cdf(-tc, df, nc)

n = 2
while power_ind(0.5, n) < 0.80:
    n += 1
print('n per group for d=0.5, power 0.80:', n)
```

```
print('power with n=20 per group:', round(power_ind(0.5, 20), 3))
# output:
# n per group for d=0.5, power 0.80: 64
# power with n=20 per group: 0.338
```

14. Excel Master Sheet

every function you need, and when to use which

Function dictionary

Function	What it gives
AVERAGE(r)	sample mean
STDEV.S(r)	sample SD (n−1) [STDEV.P divides by n: WRONG for t-tests]
VAR.S(r)	sample variance
COUNT(r)	n
DEVSQ(r)	sum of squared deviations SS
T.TEST(r1,r2,tails,type)	p-value ONLY. type 1 paired, 2 pooled, 3 Welch
T.DIST.2T(x,df)	two-tailed p from |t|
T.DIST.RT(x,df)	right-tail p
T.DIST(x,df,TRUE)	left-tail p (cumulative)
T.INV.2T(α,df)	two-tailed critical value
T.INV(1−α,df)	one-tailed (right) critical value
CONFIDENCE.T(α,s,n)	margin of error = t-crit × s/√n
F.TEST(r1,r2)	p-value for equal variances (two-tailed)

> **TIP**
> Old names TTEST, TDIST, TINV still work but are legacy. Prefer the dotted versions.

Layout A: raw data in columns

```
One-sample : data in A2:A21, claim in D1
   t   =(AVERAGE(A2:A21)-D1)/(STDEV.S(A2:A21)/SQRT(COUNT(A2:A21)))
   p   =T.DIST.2T(ABS(t),COUNT(A2:A21)-1)

Pooled     : A2:A9 and B2:B9
   p   =T.TEST(A2:A9,B2:B9,2,2)
Welch      : p =T.TEST(A2:A9,B2:B9,2,3)
```

```
Paired     : p =T.TEST(A2:A9,B2:B9,2,1)

Variance check: =F.TEST(A2:A9,B2:B9)   (p > 0.05 -> pooled ok)
```

Layout B: only summary statistics

Put $mean_1$, SD_1, n_1 in B1:B3 and $mean_2$, SD_2, n_2 in B4:B6.

```
Pooled:
B8   pooled var =((B3-1)*B2^2+(B6-1)*B5^2)/(B3+B6-2)
B9   SE         =SQRT(B8*(1/B3+1/B6))
B10  t          =(B1-B4)/B9
B11  df         =B3+B6-2
B12  p          =T.DIST.2T(ABS(B10),B11)

Welch:
B14  SE         =SQRT(B2^2/B3+B5^2/B6)
B15  t          =(B1-B4)/B14
B16  df         =(B2^2/B3+B5^2/B6)^2/((B2^2/B3)^2/(B3-1)
                  +(B5^2/B6)^2/(B6-1))
B17  p          =T.DIST.2T(ABS(B15),B16)
```

WATCH OUT

T.DIST.2T accepts a non-integer df, but older Excel versions TRUNCATE it.

For a Welch df like 17.72 the difference is tiny, but do not round yourself.

Decision cell

```
=IF(p<0.05,"Reject H0: significant","Fail to reject H0")
=IF(ABS(t)>T.INV.2T(0.05,df),"Reject H0","Fail to reject H0")
```

ToolPak menu cheat

Data Analysis option	Test
t-Test: Paired Two Sample for Means	Paired
t-Test: Two-Sample Assuming Equal Variances	Pooled
t-Test: Two-Sample Assuming Unequal Variances	Welch
F-Test Two-Sample for Variances	variance check
(no one-sample option)	build with formulas above

15. Python Master Code

write the tests yourself, then confirm with SciPy

Cheat list

everything in 8 lines

```
from scipy import stats
stats.ttest_1samp(x, popmean)                # one-sample
stats.ttest_ind(x, y, equal_var=True)        # pooled
stats.ttest_ind(x, y, equal_var=False)       # Welch
stats.ttest_rel(x, y)                        # paired (x - y)
stats.ttest_ind_from_stats(m1,s1,n1,m2,s2,n2, equal_var=...)
# add alternative='greater' | 'less' | 'two-sided' (default)
# result: .statistic  .pvalue  .df  .confidence_interval(0.95)
```

From-scratch versions (understand every line)

ttools.py

```
import numpy as np
from scipy import stats

def decide(t, df, alpha=0.05, tail='two'):
    if tail == 'two':
        p = 2 * stats.t.sf(abs(t), df)
        crit = stats.t.ppf(1 - alpha / 2, df)
        reject = abs(t) > crit
    elif tail == 'greater':
        p = stats.t.sf(t, df)
        crit = stats.t.ppf(1 - alpha, df)
        reject = t > crit
    else:                                    # 'less'
        p = stats.t.cdf(t, df)
        crit = stats.t.ppf(alpha, df)
        reject = t < crit
    r = lambda v, k: round(float(v), k)
    return dict(t=r(t, 4), df=r(df, 3), p=r(p, 5),
                crit=r(crit, 4), reject=bool(reject))

def one_sample(x, mu0, **kw):
    x = np.asarray(x, float)
    n = len(x)
    t = (x.mean() - mu0) / (x.std(ddof=1) / np.sqrt(n))
    return decide(t, n - 1, **kw)
```

```
def pooled(x, y, **kw):
    x, y = np.asarray(x, float), np.asarray(y, float)
    n1, n2 = len(x), len(y)
    sp2 = ((n1 - 1) * x.var(ddof=1)
           + (n2 - 1) * y.var(ddof=1)) / (n1 + n2 - 2)
    se = np.sqrt(sp2 * (1 / n1 + 1 / n2))
    return decide((x.mean() - y.mean()) / se, n1 + n2 - 2, **kw)

def welch(x, y, **kw):
    x, y = np.asarray(x, float), np.asarray(y, float)
    n1, n2 = len(x), len(y)
    a, b = x.var(ddof=1) / n1, y.var(ddof=1) / n2
    df = (a + b) ** 2 / (a ** 2 / (n1 - 1) + b ** 2 / (n2 - 1))
    return decide((x.mean() - y.mean()) / np.sqrt(a + b), df, **kw)

def paired(x, y, **kw):
    d = np.asarray(x, float) - np.asarray(y, float)
    return one_sample(d, 0, **kw)
```

Test the tools

```
A = [78, 85, 82, 90, 74, 88, 81, 79]
B = [72, 80, 75, 83, 70, 77, 74, 69]
print(pooled(A, B))
print(welch(A, B))
print(paired(A, B))
print(one_sample(A, 75, tail='greater'))
# cross-check against SciPy
print(round(stats.ttest_ind(A, B).statistic, 4))
print(round(stats.ttest_ind(A, B, equal_var=False).pvalue, 5))
# output:
# {'t': 2.799, 'df': 14.0, 'p': 0.01421, 'crit': 2.1448, 'reject': Tru
# e}
# {'t': 2.799, 'df': 13.872, 'p': 0.01431, 'crit': 2.1466, 'reject': T
# rue}
# {'t': 8.5515, 'df': 7.0, 'p': 6e-05, 'crit': 2.3646, 'reject': True}
# {'t': 3.7808, 'df': 7.0, 'p': 0.00344, 'crit': 1.8946, 'reject': Tru
# e}
# 2.799
# 0.01431
```

TIP

The from-scratch t and p equal SciPy's to 4 decimals. Writing it once teaches you what ttest_ind hides.

Pick the right call, automatically

```
def auto_test(x, y, paired_data=False, alpha=0.05):
    if paired_data:
        return stats.ttest_rel(x, y)
    p_lev = stats.levene(x, y).pvalue
    equal = p_lev > alpha                    # crude screen
    print('Levene p =', round(p_lev, 3), '->',
          'pooled' if equal else 'Welch')
    return stats.ttest_ind(x, y, equal_var=equal)
```

NOTE

Screening with Levene and then choosing the test is common but debated. Many analysts just use Welch every time. Both are defensible; say which you did.

Plot the evidence

```
import matplotlib.pyplot as plt
plt.boxplot([A, B], labels=['A', 'B'])
plt.scatter([1] * len(A), A, alpha=.6)
plt.scatter([2] * len(B), B, alpha=.6)
plt.ylabel('marks'); plt.title('Method A vs B'); plt.show()
```

16. Bonus: PyTorch and TensorFlow

the t-statistic is just tensor math

WHY THIS EXISTS

Core PyTorch and TensorFlow have no t-test function. But t is mean, variance and a square root, so we compute it with tensors, and take the CDF from SciPy. Typical use: compare two models over several seeds or folds with a PAIRED test.

Welch t-statistic in PyTorch

PyTorch

```
import torch
from scipy import stats

def welch_t_torch(x, y):
    x = torch.as_tensor(x, dtype=torch.float64)
    y = torch.as_tensor(y, dtype=torch.float64)
    n1, n2 = x.numel(), y.numel()
    a = x.var() / n1          # .var() is unbiased (n-1) by default
    b = y.var() / n2
    t = (x.mean() - y.mean()) / torch.sqrt(a + b)
    df = (a + b) ** 2 / (a ** 2 / (n1 - 1) + b ** 2 / (n2 - 1))
    p = 2 * stats.t.sf(abs(t.item()), df.item())
    return t.item(), df.item(), p

def paired_t_torch(x, y):
    d = torch.as_tensor(x, dtype=torch.float64) \
        - torch.as_tensor(y, dtype=torch.float64)
    n = d.numel()
    t = d.mean() / (d.std() / n ** 0.5)
    p = 2 * stats.t.sf(abs(t.item()), n - 1)
    return t.item(), n - 1, p
```

Same thing in TensorFlow

TensorFlow

```
import tensorflow as tf
from scipy import stats

def welch_t_tf(x, y):
```

```
    x = tf.constant(x, dtype=tf.float64)
    y = tf.constant(y, dtype=tf.float64)
    n1 = tf.cast(tf.size(x), tf.float64)
    n2 = tf.cast(tf.size(y), tf.float64)
    v1 = tf.math.reduce_variance(x) * n1 / (n1 - 1)  # TF var uses n
    v2 = tf.math.reduce_variance(y) * n2 / (n2 - 1)
    a, b = v1 / n1, v2 / n2
    t = (tf.reduce_mean(x) - tf.reduce_mean(y)) / tf.sqrt(a + b)
    df = (a + b) ** 2 / (a ** 2 / (n1 - 1) + b ** 2 / (n2 - 1))
    p = 2 * stats.t.sf(abs(float(t)), float(df))
    return float(t), float(df), p

def paired_t_tf(x, y):
    d = tf.constant(x, tf.float64) - tf.constant(y, tf.float64)
    n = float(tf.size(d))
    sd = tf.math.reduce_std(d) * (n / (n - 1)) ** 0.5   # n-1 SD
    t = float(tf.reduce_mean(d) / (sd / n ** 0.5))
    return t, n - 1, 2 * stats.t.sf(abs(t), n - 1)
```

Use case: is model A really better than model B?

```
acc_a = [0.912, 0.905, 0.921, 0.918, 0.909, 0.915, 0.923, 0.910]
acc_b = [0.901, 0.899, 0.915, 0.908, 0.903, 0.907, 0.912, 0.902]
print(paired_t_torch(acc_a, acc_b))      # or paired_t_tf(...)
# expected (checked with SciPy ttest_rel):
#   t = 10.667, df = 7, p = 0.00001
print(welch_t_torch([98,104,101,95,103,99,107,100,96,102],
                    [92,140,118,125,96,131]))
# expected: t = -2.075, df = 5.22, p = 0.0903
```

ML-specific warnings

- Same seed/fold for both models → the readings are PAIRED. Use the paired test.
- Cross-validation folds share training data, so scores are NOT independent. The plain paired t-test is over-confident. Use the Nadeau–Bengio corrected resampled t-test, which inflates the variance by a factor $(1/k + n_{test}/n_{train})$.
- Test on many random seeds, not one lucky run. Ten or more seeds is a good habit.
- A significant t with a tiny gain (0.2% accuracy) may not matter in practice. Report d and CI.

INTUITION

Intuition for ML folks: t is a signal-to-noise ratio of the score gap. Gap = mean improvement. Noise = how much that improvement jumps between seeds.

17. Common Mistakes and Exam Traps

what costs marks, and what costs credibility

Top mistakes

1. Using an independent test on paired data (or the reverse). Check the DESIGN first.
2. Saying 'accept H_0'. Correct: 'fail to reject H_0'.
3. Reading p as 'probability H_0 is true'. It is P(data this extreme | H_0).
4. Using STDEV.P (divide by n) instead of STDEV.S (n − 1).
5. Forgetting the $\sqrt{n}$: dividing by s instead of $s/\sqrt{n}$ in the one-sample test.
6. Wrong df: pooled is n_1+n_2-2, not n−1. Paired uses PAIRS.
7. Rounding Welch df to an integer before asking the software for p.
8. Choosing one-tailed AFTER seeing which way the data went.
9. Mixing up t-crit for one-tailed and two-tailed (0.05 one-tail = 0.10 two-tail column).
10. Assuming equal variance because it is 'standard'. Check SD ratio or just use Welch.
11. Running many t-tests across 3+ groups. Use ANOVA.
12. Ignoring outliers or non-normal differences with tiny n.
13. Subtracting in an inconsistent order: d = after − before, then keep that sign everywhere.
14. Equating 'significant' with 'important'. Always report effect size and CI.
15. Concluding 'no difference' from a non-significant result with tiny n (low power).
16. Reporting t without df. Write $t(14) = 2.80$, $p = 0.014$.

Exam tricks

- Question gives SD or variance? Check! Variance is SD squared.
- 'Standard error given' means skip the $\sqrt{n}$ step.
- Large |t| but wrong sign for a one-tailed claim → FAIL to reject.
- Report format: t(df) = value, p = value, plus one sentence in the original

words.

FINAL CHECK

30-second self-check: (1) right test? (2) df right? (3) tails right? (4) sentence in plain words?

18. Formula Cheat Sheet

one-page recall + critical value table

$$\text{One-sample: } \frac{\bar{x} - \mu_0}{s/\sqrt{n}}, \quad df = n-1$$

$$\text{Pooled: } \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{[\, s_p^2(1/n_1+1/n_2)\,]}}, \quad df = n_1+n_2-2$$

$$\text{Welch: } \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{(\, s_1^2/n_1 + s_2^2/n_2\,)}}, \quad df = \text{Satterthwaite}$$

$$\text{Paired: } \frac{\bar{d}}{s_d/\sqrt{n}}, \quad df = n-1$$

Test	Excel p-value	SciPy
One-sample	T.DIST.2T(ABS(t),n-1)	ttest_1samp(x, mu0)
Pooled	T.TEST(a,b,2,2)	ttest_ind(a,b)
Welch	T.TEST(a,b,2,3)	ttest_ind(a,b,equal_var=False)
Paired	T.TEST(a,b,2,1)	ttest_rel(a,b)

t critical values

df	2-tail 10%	2-tail 5%	2-tail 1%	1-tail 5%	1-tail 1%
1	6.314	12.706	63.657	6.314	31.821
2	2.920	4.303	9.925	2.920	6.965
3	2.353	3.182	5.841	2.353	4.541
4	2.132	2.776	4.604	2.132	3.747
5	2.015	2.571	4.032	2.015	3.365
6	1.943	2.447	3.707	1.943	3.143
7	1.895	2.365	3.499	1.895	2.998
8	1.860	2.306	3.355	1.860	2.896

9	1.833	2.262	3.250	1.833	2.821
10	1.812	2.228	3.169	1.812	2.764
12	1.782	2.179	3.055	1.782	2.681
15	1.753	2.131	2.947	1.753	2.602
20	1.725	2.086	2.845	1.725	2.528
25	1.708	2.060	2.787	1.708	2.485
30	1.697	2.042	2.750	1.697	2.457
40	1.684	2.021	2.704	1.684	2.423
60	1.671	2.000	2.660	1.671	2.390
120	1.658	1.980	2.617	1.658	2.358
∞	1.645	1.960	2.576	1.645	2.326

RULE

Decision: $|t| > t\text{-crit} \rightarrow$ reject H_0. Same as $p < \alpha$.

19. EXERCISE BANK

10 problems, all four test types, mixed order

HOW TO ATTEMPT

For each: (1) pick the test, (2) state H_0 / H_1, (3) compute t and df, (4) compare with t-crit or p, (5) write the decision. Use $\alpha = 0.05$. Answer key is on the last page.

Q1 Cereal boxes

A box is labelled 500 g. Ten boxes weigh: 498, 502, 495, 489, 501, 497, 492, 500, 496, 493 g. Is the mean weight different from 500 g? (two-tailed)

Q2 Class exam score

A standard exam has historical mean 50. A class of $n = 36$ scores mean 52.4 with SD 6.2. Does the class differ from the historical mean? (two-tailed)

Q3 Battery life claim

A phone maker claims 12 hours. A test of $n = 16$ phones gives mean 11.2 h, SD 1.5 h. Is the true mean LESS than 12 h? (one-tailed, left)

Q4 Online vs classroom

Group 1 (online): $n = 15$, mean 72.5, SD 8.1. Group 2 (classroom): $n = 12$, mean 68.2, SD 7.6. Assume equal variances. Do the means differ? (two-tailed)

Q5 Daily sales at two stores

Units sold on different days. Assume equal variances. Do the means differ? (two-tailed)

Store X: 23, 25, 28, 30, 27, 26, 24, 29

Store Y: 20, 22, 24, 21, 25, 23, 19, 22

Q6 Training programme

Programme group: $n = 20$, mean 85, SD 10. Control: $n = 22$, mean 80, SD 9. Assume equal variances. Is the programme mean GREATER than control? (one-tailed, right)

Q7 Recovery time (days)

Treatment P: $n = 12$, mean 65, SD 4. Treatment Q: $n = 15$, mean 58, SD 12. The SDs are very different. Do the means differ? (two-tailed)

Q8 Fuel efficiency (km/L)

Two engine designs with very different spreads. Do the means differ? (two-tailed)

Design P: 14.2, 15.1, 13.8, 14.9, 15.4, 14.0

Design Q: 11.5, 16.8, 9.9, 18.2, 12.4, 17.5, 10.8

Q9 Blood pressure before/after a drug

Systolic BP (mmHg) of 8 patients. Did BP drop significantly? (use the two-tailed statistic)

Before: 140, 152, 138, 160, 145, 155, 148, 150

After : 132, 148, 135, 151, 140, 150, 144, 147

Q10 Typing test before/after training

The same 14 employees are tested before and after training. Mean improvement $\bar{d} = 2.1$ words, $s_d = 3.4$. Did training INCREASE speed? (one-tailed, right)

> **TIP**
>
> Stuck on the choice of test? Re-read Section 11. Ask: one group, pairs, or two unrelated groups? Then: similar variance?

20. Answer Key

check your work: test type, t-value, decision

Verify only AFTER you have attempted every problem.

Q	Test type	t-value	df	Decision
Q1	One-sample	−2.809	9	Reject H_0
Q2	One-sample	2.323	35	Reject H_0
Q3	One-sample (left-tailed)	−2.133	15	Reject H_0
Q4	Independent, equal variance	1.408	25	Fail to Reject H_0
Q5	Independent, equal variance	4.025	14	Reject H_0
Q6	Independent, equal variance (right-tailed)	1.706	40	Reject H_0
Q7	Independent, unequal variance (Welch)	2.117	17.72	Reject H_0
Q8	Independent, unequal variance (Welch)	0.515	6.48	Fail to Reject H_0
Q9	Paired	6.494	7	Reject H_0
Q10	Paired (right-tailed)	2.311	13	Reject H_0

Verification notes

- Q3 and Q10 are one-tailed. Critical values: −1.753 (df 15, left) and +1.771 (df 13, right).
- Q6 rejects only because it is one-tailed (+1.684 at df 40). Two-tailed it would fail.
- Q7 is borderline. Q8's SDs are wildly different, so pooling would be wrong.
- Q9: $\bar{d} = 5.125$, $s_d = 2.232$. Differences taken as before − after.

> **TIP**
> Got a different t? Check: SD vs variance, n vs n−1, df, sign convention, and whether you used the pooled formula where Welch was needed.

~ end of notes ~

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